Mapping Strategies for Cross-Domain Recommendation

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ABSTRACT

Cross-domain recommendation (CDR) tackles the cold-start problem by using a user's history in a source domain to recommend items in a target domain. We build a movies-to-books recommender on the Amazon Reviews data and compare three mapping strategies, shared, cluster-based and user-specific, each trained on 10%, 20%, 50% and 100% of the available overlap users.

KEYWORDS

Cross-Domain Recommendation, Cold Start, Embedding Mapping

1 INTRODUCTION

Recommender systems struggle with users who have no history in a domain. CDR addresses this by transferring knowledge from a source domain where the user is active. We take movies as the source and books as the target. Embedding-and-mapping methods first learn user embeddings in each domain, then train a mapping from source to target embeddings on overlapping users, that is, users with ratings in both domains.

2 FORMAL DEFINITION

Input and output. Let $\mathcal{U}$, $\mathcal{B}$, $\mathcal{M}$ be the sets of users, books and movies; each user u has a book history $H_b(u) \subseteq \mathcal{B}$ and a movie history $H_m(u) \subseteq \mathcal{M}$. Our solution is a function f that takes a user–book pair (u, b), where u is a cold-start user with a movie history ($H_m(u) \neq \emptyset$, $|H_b(u)| < T$), and outputs a relevance score $f(u, b) \in \mathbb{R}$; ranking candidate books by this score gives the recommendations.

Data. The raw data [4] are triples $M' = \{(u, m, r)\}$ and $B' = \{(u, b, r)\}$ with rating $r \in \{1, \ldots, 5\}$, stored as sparse user-by-item matrices R_M, R_B (missing entries are not 0). The overlap set is $\mathcal{O} = \{u \mid H_b(u) \neq \emptyset \wedge H_m(u) \neq \emptyset\}$, with true overlap ratio $\rho_{true} = |\mathcal{O}|/|\mathcal{U}|$. Before any training we fix a random test set $\mathcal{O}_{test} \subseteq \mathcal{O}$ (20%) and remove its book ratings from B', so these users are cold-start ($T = 1$) and the removed ratings are evaluation ground truth. Let $\mathcal{O}_{train} = \mathcal{O} \setminus \mathcal{O}_{test}$.

Embeddings. We factorize $R_M \approx U' V'^{\top}$ and $R_B \approx U'' V''^{\top}$ with $U', U'' \in \mathbb{R}^{|\mathcal{U}| \times k}$, $V' \in \mathbb{R}^{|\mathcal{M}| \times k}$, $V'' \in \mathbb{R}^{|\mathcal{B}| \times k}$. Rows are embeddings: $f_m(u) = U'[u]$, $f_b(u) = U''[u]$, $v_b = V''[b]$. Starting from small Gaussian noise (standard deviation 0.01), they are trained by mini-batch gradient descent on the regularized squared error over observed ratings [2]:

$$\ell_{P,Q}(u, i, r) = (r - P[u] \cdot Q[i])^2 + \lambda\big(\|P[u]\|^2 + \|Q[i]\|^2\big) \quad (1)$$

$$(U', V') = \arg\min \sum_{(u,m,r) \in M'} \ell_{U',V'}(u, m, r), \;\; (U'', V'') \text{ likewise on } B' \quad (2)$$

The dot product of a user and an item embedding predicts a rating, giving the dense representation the mappings operate on; λ discourages large embeddings that overfit. The constant 2 in the gradient is absorbed into the learning rate α. Hyperparameters: k, λ, α, batch size, epochs.

Overlap subsets. From $\mathcal{O}_{train}$ we sample without replacement $\mathcal{O}_{10}, \mathcal{O}_{20}, \mathcal{O}_{50}, \mathcal{O}_{100}$ with 10%, 20%, 50%, 100% of its users. All mappings use the same subsets; $\mathcal{O}_{test}$ is never used for training.

Mappings. A mapping $g : \mathbb{R}^k \to \mathbb{R}^k$ translates a movie embedding into the book space. For each subset $\mathcal{O}_\rho$, each strategy minimizes by gradient descent (3 strategies $\times$ 4 subsets = 12 trained mappings), following the embed-then-map idea of EMCDR [3]:

$$L_g = \sum_{u \in \mathcal{O}_\rho} \|g(f_m(u)) - f_b(u)\|^2 \quad (3)$$

Shared: one MLP for all users, $g_{shared}(x) = W_2\, \sigma(W_1 x + b_1) + b_2$, with $W_1 \in \mathbb{R}^{h \times k}$, $W_2 \in \mathbb{R}^{k \times h}$ and ReLU σ [3]; the nonlinearity captures translations a single linear map cannot. *Cluster:* K-means splits $\{f_m(u) : u \in \mathcal{O}_\rho\}$ into $C_1..C_K$ with centroids c_j; one MLP g_j of the same form is trained per cluster on its users, and a user u uses g_{j^*}, $j^* = \arg\min_j \|f_m(u) - c_j\|$ (a simplification of MacCDR [1]); it assumes users with similar movie taste translate to book taste similarly. *Specific:* a meta-network $m_\phi : \mathbb{R}^k \to \mathbb{R}^{k \times k}$ outputs a user-specific matrix, $g_{specific}(f_m(u)) = m_\phi(f_m(u))\, f_m(u)$; only ϕ is trained, so no per-user mapping is stored (a simplification of PTUPCDR [5]). Hyperparameters: h, K, width of m_ϕ, learning rate, batch size, epochs.

Scoring. For $g \in \{g_{shared}, g_{cluster}, g_{specific}\}$,

$$f(u, b) = \begin{cases} g(f_m(u)) \cdot v_b & |H_b(u)| < T \\ f_b(u) \cdot v_b & \text{otherwise.} \end{cases} \quad (4)$$

Books are ranked by descending score, and the ranking is evaluated with NDCG@n against the held-out ratings of $\mathcal{O}_{test}$ (the rating as graded relevance, unrated books as 0). Figure 1 shows the whole pipeline.

3 PSEUDOCODE

Algorithm 1 trains embeddings, Algorithm 2 runs the full experiment and Algorithm 3 recommends books to a cold-start user.

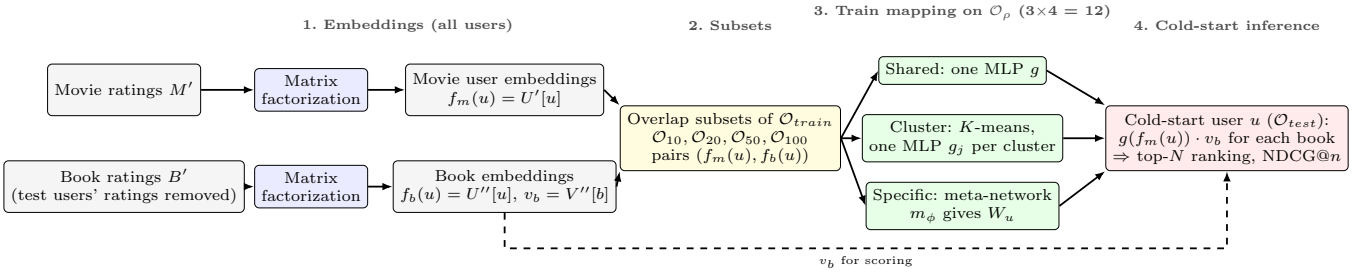


Figure 1: Pipeline: embeddings are learned per domain, mappings are trained on overlap subsets, and cold-start test users are scored through the mapping.

Algorithm 1 TrainEmbeddings (run on M' and on B')

Input: triples $D = \{(u, i, r)\}$, k, λ, α, batch size s, epochs E, seed
Output: P (users), Q (items)

set seed; $P[u], Q[i] \sim \mathcal{N}(0, 0.01^2)^k$
for $e = 1..E$; shuffle D **do**
  for each mini-batch $S \subseteq D$ of size s **do**
    $\Delta P, \Delta Q \leftarrow 0$
    for $(u, i, r) \in S$ **do**
      $err \leftarrow r - P[u] \cdot Q[i]$
      $\Delta P[u]$ += $err\, Q[i] - \lambda P[u]$; $\Delta Q[i]$ += $err\, P[u] - \lambda Q[i]$
    end for
    $P \leftarrow P + \alpha \Delta P / |S|$; $Q \leftarrow Q + \alpha \Delta Q / |S|$
  end for
end for
return P, Q

Algorithm 2 RunExperiment

Input: M', B', T=1, fractions {.1, .2, .5, 1}, K, seed
Output: NDCG@n per strategy and fraction

$\mathcal{O} \leftarrow \{u : H_b(u) \neq \emptyset \wedge H_m(u) \neq \emptyset\}$; $\mathcal{O}_{test} \leftarrow$ random 20% of $\mathcal{O}$
$truth[u] \leftarrow$ book ratings of $u \in \mathcal{O}_{test}$; remove them from B'
$(U', V') \leftarrow$ TrainEmbeddings(M'); $(U'', V'') \leftarrow$ TrainEmbeddings(B')
for each fraction ρ **do**
  $\mathcal{O}_\rho \leftarrow$ sample $\lfloor \rho |\mathcal{O} \setminus \mathcal{O}_{test}| \rfloor$ users without replacement
  $X \leftarrow \{U'[u]\}$, $Y \leftarrow \{U''[u]\}$ for $u \in \mathcal{O}_\rho$
  $g_{shared} \leftarrow$ Fit(g_θ, X, Y)
  $c_{1..K}, C_{1..K} \leftarrow$ KMeans(X, K); $g_j \leftarrow$ Fit($g_{\theta_j}, X_{C_j}, Y_{C_j}$)
  $m_\phi \leftarrow$ Fit($m_\phi(x)\, x, X, Y$)
  for each strategy g **do**
    $rank_u \leftarrow$ Recommend(u, g, T, n) for $u \in \mathcal{O}_{test}$ (models trained for ρ)
    $score_g(\rho) \leftarrow$ mean NDCG@$n(rank_u, truth[u])$
  end for
end for
Fit(g_θ, X, Y): for E epochs and each mini-batch S,
  $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{|S|} \sum_{(x,y) \in S} \| g_\theta(x) - y \|^2$

Algorithm 3 Recommend(u, g, T, n)

if $|H_b(u)| < T$ **then**
  $x \leftarrow U'[u]$
  $e \leftarrow g_{shared}(x)$ **or** $g_{j^*}(x)$ with $j^* = \arg\min_j \| x - c_j \|$ **or** $m_\phi(x)\, x$
else $e \leftarrow U''[u]$
end if
$score[b] \leftarrow e \cdot V''[b]$ for each candidate $b \notin H_b(u)$
return the n books with the highest $score[b]$

4 IMPLEMENTATION

Code, README and a quickstart script are at https://github.com/rishon-r/SENG474-Cross-Domain-Recommendation-System. Every random step (embedding initialization, test split, subset sampling, K-means, MLP initialization, batch order) uses one fixed seed, so a run with the same data, code and configuration reproduces the same results.

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